

Name: _____

You must show all work to receive full credit.

Answers and assertions must be fully justified.

Your solutions must be clear, concise, and easy to follow.

Your solutions must be the result of your own efforts.

1. Definitions and Theorems	
2. Short Answer and Computations	
3. Explanations	
4. Proof	
TOTAL	
5. Group	
6. Subgroup	

1. (12 points) Definitions: Complete the following definitions.

(a) The order of a finite group G is:

(b) An even permutation is:

(c) The center of a group, $Z(G)$ is:

2. (8 points) Theorems: Complete the following theorems.

(a) *Let $G = \langle a \rangle$ be a finite cyclic group of order n . Then for any integer m , $|a^m| =$*

(b) *Let H be a finite subset of a group G . Then, the finite subgroup test says that H is a subgroup of G if*

3. (65 points, 5 points each) Short Answer: For full credit, you must justify your answer by providing clear work or a very brief explanation.

(a) How many generators does $G = \langle a \rangle$ have, given that $|a| = 12$?

(b) Which element in D_4 is the same as $(R_{90})^{90}$.

(c) What is the order of the permutation $(1235)(24567)$?

(d) Given that a is a group element and $a^{15} = a^4$, express a^{-1} as a positive power of a .

(e) What is the inverse of 7 in $U(20)$?

(f) Is the permutation $(1243)(3521)(123)$ even or odd?

- (g) How many elements of order 4 are there in S_7 ?
- (h) What is the order of 8 in $\mathbb{Z}_{30}$?
- (i) Determine all subgroups of $\mathbb{Z}_{10}$.
- (j) How many subgroups does $\mathbb{Z}_{p^n}$ have, where p is a prime?
- (k) Find all elements k in $U(9)$ such that $U(9) = \langle k \rangle$.
- (l) In D_4 , what is $C(R_{180})$?
- (m) Determine the rotational or symmetry group of the design below.



4. (24 points, 6 points each) Explanations: Answer each of the following in no more than two sentences.

(a) Give two reasons why the set $K = \{2n + 1 : n \in \mathbb{Z}\}$ is not a group under addition.

(b) Suppose a cyclic group G has exactly three subgroups: G itself, $\{e\}$, and a subgroup of order 7. What can you say about $|G|$?

(c) Let G be a group. Explain why $Z(G) = \bigcap_{a \in G} C(a)$. (Note that $\bigcap_{a \in G} C(a)$ means the intersection of *all* subgroups of the form $C(a)$, for all $a \in G$.)

(d) Let $\alpha \in S_n$ have odd order. Why must α be an even permutation?

5. (12 points) Let G be a group and let $a \in G$. An element $b \in G$ is called a *conjugate of a* if there exists an element $x \in G$ such that $b = xax^{-1}$. Prove that any conjugate of a has the same order as a .

6. **Group Proof:** Define an operation $*$ on the set of integers by $a * b = a + b - 1$. Prove that under this operation, the set of integers is a group.

7. **Subgroup Proof:** Let G be an Abelian group, and define $H = \{a \in G : a^2 = e\}$. Prove that H is a subgroup of G .